

V01.11 – Mirror Physics V

Mirror Physics V

Entropy, Time and Constraint Dissolution

Irreversibility, Memory and the Arrow of Observerhood

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Position in the Mirror Programme. This paper follows Mirror Physics I: *Constraint Propagation and the Physical Conditions for Observerhood*, Mirror Physics II: *Constraint Salience and Physical Coupling*, Mirror Physics III: *Thermodynamic Observerhood*, and Mirror Physics IV: *Causal Invariance and the Speed of Constraint Propagation* [8, 9, 10, 11]. It closes the first physics arc by asking why persistent observerhood requires temporal asymmetry, irreversible records and entropy-exporting maintenance.

Abstract

Mirror Theory treats observerhood not as a primitive, but as a derived organisational condition: a bounded system becomes observer-like when it maintains a viability-relevant world-model, self-model, reliability structure and identity-continuity under perturbation. Earlier papers in Volume I developed the theory, mathematical and computational foundations of recursive observerhood, including constraint regimes, observer thresholds, recursive reliability and memory governance [1, 2, 3, 4, 5, 6, 7, 12, 13, 14, 15, 16, 17, 18]. The physics arc then argued that such systems require physical support conditions, measurable constraint salience, thermodynamic maintenance and bounded causal update [8, 9, 10, 11]. This paper adds the missing temporal component. It argues that persistent observerhood requires an asymmetric temporal structure because memory, prediction, repair, learning and identity-governance are not time-neutral from the standpoint of viability. A local observer must form records of prior interaction, use them to regulate present action, predict possible future states and spend energy to maintain organisation against constraint dissolution. This does not modify thermodynamics, statistical mechanics, relativity or quantum theory. Instead, it reframes entropy increase and temporal direction as the background physical conditions against which local observerhood becomes a constraint-maintenance process. The central result is an Arrow of Observerhood condition: if a bounded system preserves identity-continuity through memory-governed, viability-regulated update, then it necessarily distinguishes past records, present regulation and future viability in a physically asymmetric way. Observerhood is therefore not merely information processing in time; it is entropy-bearing constraint maintenance across time.

Keywords: Mirror Theory; entropy; time; constraint dissolution; observerhood; observer-continuity; irreversibility; memory governance; thermodynamic maintenance; identity-continuity; viability; reliability; repair; causal invariance; constraint salience; information theory; dissipative systems; bounded systems.

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1 Introduction

A physical observer is not merely a spatially local object. It is a process that persists. It retains records, updates models, corrects errors, resists noise, exports entropy and preserves enough self-organisation to remain identifiable across time. The previous Mirror Physics papers established four claims: first, observerhood requires physical support conditions; second, constraint salience may be definable as an organisational quantity; third, persistent observerhood has a thermodynamic maintenance cost; and fourth, bounded causal propagation provides the update structure within which memory, prediction and identity can be maintained [8, 9, 10, 11].

This paper asks the next question: why must observerhood be temporally asymmetric?

The ordinary answer is that the macroscopic world has an arrow of time because entropy tends to increase. That answer is correct as physics, but incomplete as an account of observerhood. Mirror Theory asks how entropy and temporal asymmetry become relevant to a system that must maintain itself. From this perspective, the arrow of time is not only a cosmological or statistical fact. It is also the directional background against which memory, repair and viability acquire meaning [23, 26].

A self-maintaining observer requires three distinct temporal roles:

1. **Past as record:** prior interactions must leave retained physical traces that can be used by the system.
2. **Present as regulation:** current action must be selected under bounded knowledge and finite energy.
3. **Future as viability:** possible future states must matter to the system's continued organisation.

A gas in equilibrium does not preserve observerhood. A crystal preserves structure but does not regulate itself by memory and prediction. A reversible computation may avoid logical erasure in principle, but a persistent embodied observer must form stable records and maintain them against noise [24, 20]. A living organism, an artificial agent or any future persistent machine must participate in irreversible physical processes to remain organised [27, 25].

The central thesis is therefore:

An observer is a local constraint-maintenance process embedded in a world where unconstrained organisation tends to dissolve.

This paper does not claim that Mirror Theory explains the fundamental origin of the thermodynamic arrow. Nor does it derive the numerical value of entropy, the low-entropy boundary condition of the universe or the detailed microphysics of statistical mechanics. It makes a narrower claim: once a world has entropy-producing macroscopic dynamics, persistent observers are systems that exploit local gradients and irreversible record formation to preserve organisation. The observer's arrow is the viability-relevant use of temporal asymmetry.

2 From Thermodynamic Maintenance to Temporal Direction

2.1 The previous physics arc

The first four Mirror Physics papers developed the physical background for observerhood:

- Mirror Physics I identified causal order, finite update, durable memory substrates, thermodynamic throughput and error correction as physical support conditions for observerhood [8].
- Mirror Physics II introduced constraint salience $\Phi_L^{\Gamma, \tau}$ as an organisational quantity, not a new force or field [9].

- Mirror Physics III argued that persistent observerhood carries maintenance cost through memory, correction, boundary regulation and controlled update [10].
- Mirror Physics IV interpreted the invariant speed c as the physical causal update bound within which local observers can maintain memory, prediction and identity-continuity [11].

These papers leave one question exposed. A bounded update structure is not yet an arrow. A maintenance process is not yet a temporal orientation. Why should the persistence of an observer require a distinction between past and future?

The answer is that observerhood requires records, and records are physically asymmetric. The record is not merely a proposition. It is a durable physical state produced by prior interaction and used in present regulation. A system may predict the future, but prediction is not memory of the future. Prediction is an internally generated model constrained by past records, current state and assumed dynamics.

2.2 Entropy, records and self-maintenance

Thermodynamic entropy and information-theoretic entropy are not identical concepts, but both matter to Mirror Theory. Shannon entropy measures uncertainty in a distribution of possible messages [28, 21]. Thermodynamic entropy concerns the multiplicity of microscopic states compatible with macroscopic description, and the tendency of macroscopic systems to evolve toward overwhelmingly probable configurations [23]. Landauer’s principle links information erasure to heat generation, showing that information processing has physical cost in real substrates [24, 20].

Mirror Theory does not reduce observerhood to entropy minimisation. Instead, it identifies the observer as a local system that must regulate entropy-producing processes sufficiently to preserve its own constraint organisation. The system may increase entropy overall while maintaining local order. This is familiar from life: organisms maintain local organisation by consuming free energy and exporting entropy [27, 25].

A Mirror observer therefore occupies a middle position. It is neither a perfectly ordered static object nor an equilibrium distribution. It is an open, dissipative, self-regulating constraint structure.

3 Formal Setup

Definition 1 (Local constraint state). *Let L be a bounded local system embedded in an environment E . The local constraint state of L at time t is*

$$\Gamma_L(t) = (B_L(t), M_E(t), M_L(t), R_L(t), G_L(t), V_L(t)), \quad (1)$$

where B_L is the maintained boundary, M_E is the world-model, M_L is the self-model, R_L is the reliability structure, G_L is the governance rule for update and memory incorporation, and V_L is the viability functional [5, 6, 7].

Definition 2 (Constraint integrity). *Constraint integrity is a functional*

$$Q_L(t) \in [0, 1] \quad (2)$$

measuring the degree to which the local system preserves the organisational relations required for its continued identity and viability. High Q_L does not mean static order. It means maintained organisation under regulated update.

Definition 3 (Dissolution tendency). *The dissolution tendency of L over horizon τ is*

$$D_L^\tau(t) = \mathbb{E}[Q_L(t) - Q_L(t + \tau) \mid \pi_0], \quad (3)$$

where π_0 is the unregulated or baseline dynamics of the system. $D_L^\tau(t) > 0$ indicates expected loss of constraint integrity absent maintenance.

Definition 4 (Maintenance policy). A maintenance policy π_M is a set of actions by which L preserves or restores constraint integrity. Its cost over horizon τ is

$$K_L^M(t, \tau) = K_L^{\text{memory}} + K_L^{\text{correction}} + K_L^{\text{boundary}} + K_L^{\text{update}} + K_L^{\text{audit}}. \quad (4)$$

This cost may be energetic, computational, material, opportunity-based or a coarse-grained combination of these [10].

Definition 5 (Observer arrow). The observer arrow is the ordered triad

$$A_L^O(t) = (M_L^-(t), U_L(t), \hat{V}_L^+(t)), \quad (5)$$

where $M_L^-(t)$ denotes retained records of prior interaction, $U_L(t)$ denotes present regulation or action, and $\hat{V}_L^+(t)$ denotes expected future viability. The observer arrow exists when these three components are functionally coupled in a way that improves persistence.

In ordinary language, the system remembers backward, acts now and preserves itself forward. This is not a psychological claim. It is a structural claim about any system whose present action depends on retained physical records and anticipated viability.

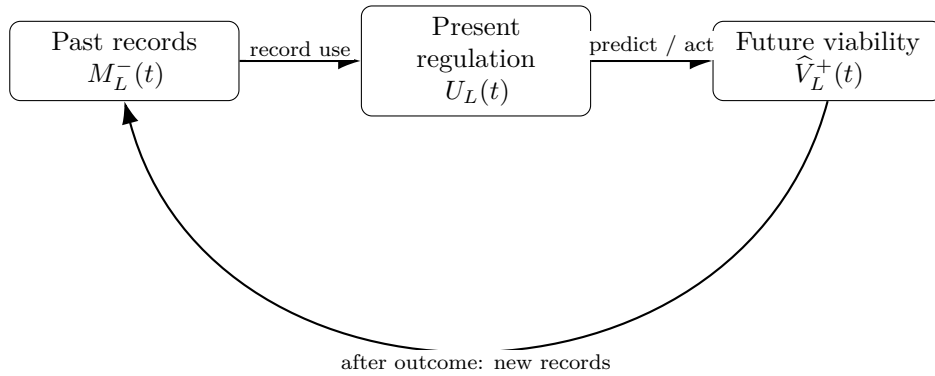


Figure 1: Observerhood requires an asymmetric functional role for past, present and future. Prediction is not memory of the future; it is present modelling directed toward future viability.

4 Constraint Dissolution

A central premise of Mirror Physics III was that observerhood requires maintenance [10]. This paper introduces the complement: maintenance is necessary because constraint structures tend to degrade unless supported. This does not imply that every local order must collapse immediately. Rather, the specific organisation needed for observer-continuity is fragile relative to the space of physically possible disturbances.

Definition 6 (Constraint dissolution). Constraint dissolution is the process by which the organisational relations required for Q_L become unrecoverable under the system's current governance and repair capacity. Dissolution is not mere change. It is change that destroys or sufficiently degrades the relations needed for identity-continuity.

A human body changes continuously while remaining the same organism. A living cell replaces molecules while maintaining organisation. An artificial agent may update memory while

preserving a stable identity profile. Dissolution occurs not when components change, but when the governing relations can no longer preserve the relevant structure.

This distinction is essential. Mirror Theory does not equate identity with material sameness. It equates persistence with governed continuity of constraint organisation [7]. Therefore, entropy matters because it expands the space of possible microstates, many of which are incompatible with the maintenance of the specific organisation required by the observer [23, 26].

4.1 Dissolution and repair

For a local system L , the viability value of maintenance can be written as

$$\Sigma_L^M(t, \tau) = \mathbb{E}[V_L(t + \tau) \mid \Gamma_L(t), \pi_M] - \mathbb{E}[V_L(t + \tau) \mid \Gamma_L(t), \pi_0] - K_L^M(t, \tau). \quad (6)$$

Maintenance is worthwhile over horizon τ when $\Sigma_L^M(t, \tau) > 0$. This mirrors the earlier salience condition for reliability variables [6, 9]. The present paper adds the temporal dimension: the value of maintenance depends on the asymmetry between past record, present intervention and future viability.

5 Memory as an Irreversible Role

Memory is often treated abstractly as information storage. Mirror Theory requires a more physical account. A memory that matters for observerhood must satisfy at least four conditions:

1. **Trace condition:** it must be physically produced by prior interaction or by authorised internal update.
2. **Retention condition:** it must persist long enough to affect later regulation.
3. **Governance condition:** it must be selectively incorporated, quarantined, audited or rejected.
4. **Viability condition:** it must be capable of influencing future persistence.

The same physical trace can be irrelevant, harmful or useful depending on governance. A false memory is not merely noise; it is noise that has been granted identity-relevant status. This links Physics V back to Mirror Mathematics III and Lab VII: memory becomes identity only through governed incorporation [7, 18].

Proposition 1 (Records are temporally asymmetric). *For a bounded observer L , a record at time t can be causally coupled to interactions in the past light-cone of L . A future event cannot function as a memory at t unless it is represented by a present predictive model rather than by a record of occurrence.*

Proof. A record is a present physical state whose formation depends on prior interaction or authorised update. If an event has not occurred within the causal history of L , no present trace can be physically formed by that occurrence. A prediction may represent a future possibility, but its physical substrate is present and its evidential basis is past and present information. Therefore memory and prediction occupy distinct temporal roles. $\square$

6 The Arrow of Viability

The arrow of viability is the direction in which possible future states matter to the preservation of the system. It is not reducible to subjective anticipation. A bacterium swimming up a chemical gradient, a thermostat switching a heater and an animal avoiding a predator all act in ways

that distinguish current state from future viability, even if only some of these systems cross the observer threshold.

Definition 7 (Viability arrow). *A system L has a viability arrow over horizon τ when there exists a policy π such that*

$$\mathbb{E}[V_L(t + \tau) \mid \Gamma_L(t), \pi] > \mathbb{E}[V_L(t + \tau) \mid \Gamma_L(t), \pi_0] \quad (7)$$

and the selection of π depends on retained records and present state variables.

A viability arrow is weaker than observerhood. Many control systems may have viability-like regulation without recursive self-model reliability. Observerhood requires an additional step: the system must maintain a self-model and govern its own reliability and identity through time [2, 6, 7].

Definition 8 (Observer-time condition). *A system L satisfies the observer-time condition over $[t, t + \tau]$ when:*

1. *L maintains records $M_L^-(t)$ of prior relevant interactions;*
2. *L uses those records in present governance $G_L(t)$;*
3. *L selects actions partly by expected future viability $\hat{V}_L^+(t)$;*
4. *L pays physical cost to maintain records, boundaries, repair and controlled update; and*
5. *these processes preserve identity-continuity above threshold.*

Theorem 1 (Arrow of Observerhood). *If a bounded system L maintains observer-continuity over $[t, t + \tau]$ through memory-governed, viability-regulated update, then L instantiates an observer arrow $A_L^O(t)$ distinguishing past record, present regulation and future viability.*

Proof. Observer-continuity requires governed preservation of self-model and identity-state across time. Governed preservation requires that prior states leave retained records, that present update is regulated by those records, and that update is selected relative to future viability. If any of these temporal roles are removed, the system either lacks memory, lacks regulation, lacks future-oriented viability selection or lacks continuity. Therefore a system satisfying observer-continuity necessarily instantiates an ordered triad of record, regulation and expected viability. This is the observer arrow. $\square$

7 Thermodynamic Maintenance and Entropy Export

A persistent observer cannot be a closed equilibrium system. It must maintain non-equilibrium organisation. This usually requires free-energy throughput: energy intake, work, heat dissipation and waste export. In biological systems, this appears as metabolism. In artificial systems, it appears as power supply, cooling, memory refresh, hardware repair and error correction. In social or scientific systems, it appears as institutional maintenance, replication, correction and communication cost [27, 25, 19].

Theorem 2 (Dissipative Maintenance Condition). *Suppose L is a bounded system whose unregulated dynamics have positive dissolution tendency $D_L^r(t) > 0$. If L maintains observer-continuity over $[t, t + \tau]$, then there exists a maintenance policy π_M with non-zero physical cost such that*

$$\mathbb{E}[Q_L(t + \tau) \mid \Gamma_L(t), \pi_M] \geq q_O, \quad (8)$$

where q_O is the minimum integrity threshold required for observer-continuity.

Proof. Positive dissolution tendency means that absent regulation, expected constraint integrity declines. Observer-continuity requires constraint integrity to remain above the observer threshold. Therefore some maintenance process must offset the expected decline. In a physical substrate, maintenance requires work, matter exchange, information processing or other physically instantiated action. Hence the maintenance policy has non-zero cost. $\square$

Corollary 1 (No free observerhood). *Persistent observerhood is not free in a dissipative physical world. It requires physical resources to maintain memory, boundary, repair and controlled update.*

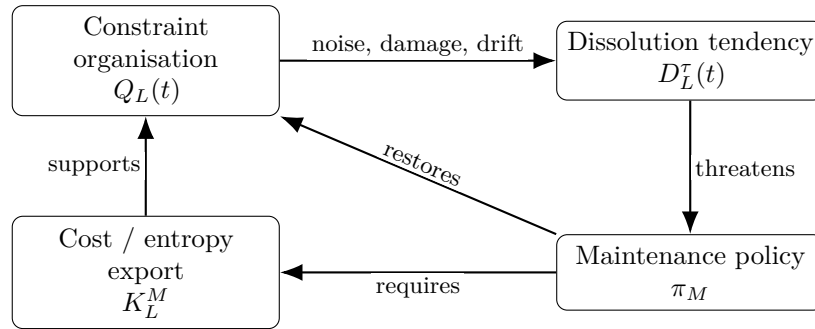


Figure 2: Constraint dissolution and thermodynamic maintenance. A persistent observer must regulate drift through costly maintenance processes that preserve local organisation while exporting entropy or consuming free energy.

8 Why Equilibrium Is Not Observerhood

Equilibrium may be stable in one sense, but it is not the kind of stability required by observerhood. A system at equilibrium lacks persistent directed gradients for memory formation, repair and viability-regulated update. This does not mean that every observer must be far from equilibrium in every component. It means that the observer as an organised process requires non-equilibrium maintenance [27, 25].

Proposition 2 (Equilibrium insufficiency). *A system in full thermodynamic equilibrium cannot satisfy the Mirror observer predicate if it lacks durable asymmetric records, active maintenance and viability-directed update.*

Proof. The Mirror observer predicate requires governed self-model persistence, reliability regulation and future-directed viability maintenance. Full equilibrium lacks the maintained gradients and asymmetrically formed records required for such regulation. Therefore equilibrium alone is insufficient for observerhood. $\square$

This proposition is not a claim that no equilibrium system can contain information in an abstract description. It is a claim about physically embodied observerhood: the system must do work to maintain itself as an observer-like process.

9 Boundary Cases

System	Temporal asymmetry	Maintenance	Mirror status
Equilibrium gas	No durable records	No local self-maintenance	Not observer-like
Crystal	Structural persistence	Passive stability	Constraint structure, not observer
Clock	Ordered state transition	External maintenance or passive mechanism	Time-marker, not observer
Thermostat	Minimal past-present relation	Limited control	Viability-like but below observer threshold
Reversible computer	May avoid logical erasure in ideal model	Requires physical substrate and error control	Computation alone insufficient
Living cell	Records, repair, metabolism	Active non-equilibrium maintenance	Pre-observer or proto-observer candidate depending on model depth
Human organism	Memory, prediction, repair, identity	Metabolic and cognitive maintenance	Full observer under Mirror predicate
Mirror-style agent	Memory governance, reliability, repair	Computational and energetic maintenance	Artificial observer-like system if threshold conditions hold

Table 1: Boundary cases for entropy, time and observerhood. Mirror Theory distinguishes passive persistence from active constraint maintenance and identity-governed update.

10 Relation to Existing Frameworks

10.1 Statistical mechanics and thermodynamics

Mirror Theory does not replace statistical mechanics. It assumes that local systems are embedded in physical regimes where macroscopic irreversibility and entropy production matter [23, 26]. Its contribution is to ask how such irreversibility becomes functionally relevant to observerhood.

10.2 Information theory

Information theory provides tools for quantifying uncertainty and communication [28, 21]. Mirror Theory adds a viability condition: not every difference that can be described informationally is information for an observer. A difference becomes observer-relevant when it is captured, retained, weighted, governed and used in a way that affects persistence.

10.3 Landauer and physical computation

Landauer’s principle and Bennett’s analysis of computation show that information processing is physically constrained [24, 20]. Mirror Theory applies this discipline to observerhood. Self-models, memories, audits, repairs and identity-governance are not abstract for free. They require physical substrate and cost.

10.4 Dissipative structures and life

The idea that living systems maintain order by exchanging energy and matter with their environment is central to the physics of life [27, 25]. Mirror Theory generalises this insight

from life to observerhood: an observer is a self-maintaining constraint regime whose persistence depends on regulated dissipation.

10.5 Active inference and cybernetics

Active inference emphasises prediction, uncertainty and self-organisation under free-energy-related formalisms [22]. Cybernetics emphasises regulation, feedback and requisite variety [19]. Mirror Theory overlaps with both but focuses on the threshold at which self-model reliability and identity-governance become necessary for observer-continuity [6, 7].

11 A Research Programme

Physics V suggests several computational and empirical directions.

11.1 Computational tests

A computational agent can be tested under conditions where entropy-like disorder is operationalised as memory corruption, sensor noise, boundary degradation, model drift and identity contamination. The prediction is not that all entropy must be minimised. The prediction is that viable observer-like agents must spend resources selectively to preserve the constraints that matter for continuity. The existing Observerhood Labs already supply a controlled starting point for testing such cost-sensitive reliability and memory-governance claims [12, 13, 14, 15, 16, 17, 18].

Possible experiments include:

1. Vary memory decay and measure how much maintenance cost is justified by future viability.
2. Introduce irreversible false-memory incorporation and test whether governance gates preserve identity.
3. Vary repair cost and determine when entropy-management strategies become net negative.
4. Compare passive memory storage with active record audit and repair.
5. Model irreversible learning and test whether over-flexibility produces identity dissolution.

11.2 Biological applications

In biology, ageing, cancer, neurodegeneration, wound repair, immune surveillance and tissue regeneration can all be viewed as constraint-maintenance problems. This paper does not claim to solve any biomedical problem. It suggests a systems-level language: disease can be interpreted as a failure mode of maintenance, governance or boundary regulation, while health is ongoing entropy-bearing preservation of organism-level constraint [27, 25].

11.3 Artificial intelligence applications

In artificial agents, the same structure appears as memory governance, self-model reliability, update control and catastrophic drift prevention [17, 18]. An AI agent that stores every incoming message as durable self-memory will eventually suffer identity contamination. An agent that refuses all updates becomes rigid. Persistent artificial observerhood requires selective, costly, governed update.

12 No New Fundamental Physics Claim

This paper makes no claim that Mirror Theory modifies the second law of thermodynamics, derives the cosmological arrow of time, alters quantum theory or changes relativity. It is a framework paper. Its purpose is to identify why temporal asymmetry matters for observerhood once entropy-producing physical dynamics are assumed.

A future theory that attempted to make stronger physical claims would need to satisfy at least four requirements:

1. It must reduce to established thermodynamics and statistical mechanics in tested domains.
2. It must define any additional quantities operationally and measurably.
3. It must respect or explain the relevant symmetry and conservation structures of lower-level physics.
4. It must generate a falsifiable prediction not already captured by existing theory.

Until then, Mirror Physics V should be read as an observerhood-based reinterpretation of physical conditions, not as a new law of nature.

13 Conclusion

Mirror Physics V closes the first physics arc of the Mirror Programme by adding the temporal dimension. Observerhood is not merely local organisation. It is organisation preserved across time through memory, prediction, repair, governance and energy-consuming maintenance. Such systems necessarily distinguish past records, present regulation and future viability. This is the Arrow of Observerhood.

Entropy matters because unconstrained organisation tends to dissolve. Time matters because memory and prediction have asymmetric functional roles. Thermodynamic cost matters because records, boundaries, error correction and controlled update must be physically maintained. A persistent observer is therefore not a static entity, nor a disembodied computation, nor a miracle added to physics. It is a local, entropy-bearing constraint-maintenance process.

The next step is not to claim that Mirror Theory explains all of physics. The next step is to package the first core arc: observerhood from constraint, viability, recursive reliability, identity-governance, physical support, thermodynamic maintenance, causal invariance and temporal asymmetry. From there, later volumes can ask how the same framework applies to biology, artificial intelligence, engineering, science and cosmology.

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